

Q-1

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a-) $2^n + 1 = O(2^n) \Rightarrow$ For check this we can use limit.

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \Rightarrow \lim_{n \rightarrow \infty} \frac{2^n + 1}{2^n} \Rightarrow \frac{2^n (2^n + \frac{1}{2^n})}{2^n} = \lim_{n \rightarrow \infty} 2^n = \infty \text{ so we can say that:}$$

$2^n + 1 = O(2^n)$ is True.

b-) $n^3 - 4n^2 + 7 = O(2^n)$ In this question we can use biggest terms for limit.

$$\lim_{n \rightarrow \infty} \left(\frac{n^3 - 4n^2 + 7}{2^n} \right) = \lim_{n \rightarrow \infty} \left(\frac{n^3}{2^n} \right)$$

In this case 2^n 's growth is much faster than n^3 's growth.

So this limit goes to zero. Therefore we cannot say that

these two expressions grow exactly the same.

So $O(2^n)$ is False. $\Rightarrow O(2^n) \checkmark$

c-) $n^3(1+\sqrt{n}) = O(n^3 \log n)$ L'opital Rule

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^3(1+\sqrt{n})}{n^3 \cdot \log n} = \frac{1+\sqrt{n}}{\log n} = \frac{1}{2\sqrt{x}} \cdot x = \frac{x}{2\sqrt{x}} = \frac{\sqrt{x}}{2} = \lim_{n \rightarrow \infty} \frac{\sqrt{x}}{2} = \infty$$

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So $O(n^3 \log n)$ is False. Correct answers: $\Omega(n^3 \log n)$ or $O(n^3 \sqrt{n}) = O(n^{3.5})$

d-) $28n = O(7n^2) \Rightarrow \lim_{n \rightarrow \infty} \frac{28n}{7n^2} = \frac{4}{n} = \lim_{n \rightarrow \infty} \left(\frac{4}{n}\right) = 0$ So $O(7n^2)$ is true

e-) $n + \log n + 21 = O(7n^2)$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n + \log n + 21}{7n^2} = \lim_{n \rightarrow \infty} \left(\frac{n}{7n^2} + \frac{\log n}{7n^2} + \frac{21}{7n^2} \right) = \lim_{n \rightarrow \infty} \left(\frac{\log n}{7n^2} \right) \xrightarrow{\text{L'opital Rule}} \frac{1}{n} \cdot \frac{1}{14n} = \frac{1}{14n^2}$$

$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{14n^2} = 0$ So $O(7n^2)$ is true

$$F.) \quad n^2 + 9n - 13 = O(n^2)$$

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$$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{n^2 + 9n - 13}{n^2} \right) = \frac{n^2 \left(1 + \frac{9}{n} - \frac{13}{n^2} \right)}{n^2} = \lim_{n \rightarrow \infty} (1) = 1 \quad \text{so } \underline{O(n^2) \text{ is True}}$$

Question-2

$$2n^2, 3n^4, 7n, \log n, 3^n, n!$$

① Compare $2n^2$ and $3n^4$ by using limit:

$$\lim_{n \rightarrow \infty} \frac{2n^2}{3n^4} = \frac{2n^2}{3 \cdot n^2 \cdot n^2} = \frac{2}{3n^2} = \lim_{n \rightarrow \infty} \frac{2}{3n^2} = 0$$

So we can say that growth rates = $2n^2 < 3n^4$

② Compare $3n^4$ and $7n$

$$\lim_{n \rightarrow \infty} \frac{3n^4}{7n} = \frac{3n^3}{7} = \lim_{n \rightarrow \infty} \frac{3n^3}{7} = \infty$$

So growth rates = $7n < 3n^4$

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③ Now compare $2n^2$ and $7n$

$$\lim_{n \rightarrow \infty} \frac{2n^2}{7n} = \lim_{n \rightarrow \infty} \frac{2n}{7} = \infty$$

So growth rates = $7n < 2n^2$

④ Compare $7n$ and $\log n$

$$\lim_{n \rightarrow \infty} \frac{7n}{\log n} = \text{using L'Hopital} = \lim_{n \rightarrow \infty} \frac{7 \cdot x}{1} = \lim_{n \rightarrow \infty} 7x = \infty$$

So growth rates = $\log n < 7n$

⑤ Compare 3^n and $3n^4$

$$\lim_{n \rightarrow \infty} \frac{3^n}{3n^4} = \text{using L'Hopital} = \frac{3^n \cdot \ln 3}{12n^3} = \text{using L'Hopital Again} = \frac{3^n \cdot (\ln 3)^2}{36n^2}$$

$$= \text{using L'Hopital Again} = \frac{3^n (\ln 3)^3}{72n} = \text{using L'Hopital Again} = \frac{3^n (\ln 3)^4}{72} = \lim_{n \rightarrow \infty} \frac{3^n (\ln 3)^4}{72}$$

= ∞ So growth rates = $3n^4 < 3^n$

Until now, if we use above results:

$$\log n < 7n < 2n^2 < 3n^4 < 3^n$$

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And we can easily say that Factorial Functions grow much faster than exponential Functions. So growth rates;

$$\log n < 7n < 2n^2 < 3n^4 < 3^n < n!$$

Question-3

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<u>Step</u>	<u>Step</u>	<u>Freq</u>
a.) int sum = 0 ; $\Rightarrow$ 1		1
for (int i = 0; i < a; i++)	i = 0 $\Rightarrow$ 1 i < a $\Rightarrow$ 1 i++ $\Rightarrow$ 1	1 a+1 a
for (int j = 0; j < b; j++)	j = 0 $\Rightarrow$ 1 j < b $\Rightarrow$ 1 j++ $\Rightarrow$ 1	a a(b+1) a.b
sum += i + b;	= $\Rightarrow$ 1 + $\Rightarrow$ 1	a x b a x b

+ _____

$$\text{Total Freq} = 3 + 3a + 4ab$$

$O(ab)$

$$\text{Complexity} = O(ab)$$

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b-) Step-1 ($i=0$)
 $i = 2 \times 0 + 1 = 1$

Step-2 ($i=1$)
 $i = 2 \times 1 + 1 = 3$

Step-3 ($i=3$)
 $i = 2 \times 3 + 1 = 7$

Step-4 ($i=7$)
 $i = 2 \times 7 + 1 = 15$
 $\vdots$

General formula is $2^n - 1$

$$2^n - 1 < a \Rightarrow 2^n \leq a + 1$$

$$\Rightarrow n = \log_2^{a+1}$$

number of cycle

int sum = 0 : Step
int sum = 0 $\Rightarrow$ 1

for (int i = 0; i < a; i++)
 $i = 0; \Rightarrow 1$
 $i < a \Rightarrow 1$
 $i++ \Rightarrow 1$

sum += 1; $\Rightarrow$ 2 (addition + assign)

$i = i * 2 \Rightarrow$ 2 (assign + mult.p.)

Freq
1

1

$\log_2^{a+1} + 1$

$\log_2^{a+1}$

$2 \cdot \log_2^{a+1}$

$2 \cdot \log_2^{a+1}$

+
Total Freq = $6 \log_2^{a+1} + 3$

Complexity = $O(\log_2^a)$

C-)

For(int i=0; i<n; i++)

Step
 $i=0 \Rightarrow 1$ $i<n \Rightarrow 1$ $i++ \Rightarrow 1$ Freq

1

 $n+1$ n Gökem Uysal
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For(int j=0; j<n; j++)

 $j=0 \Rightarrow 1$ $j<n \Rightarrow 1$ $j++ \Rightarrow 1$ n

$$n \times (n+1) = n^2 + n$$

$$n \times n = n^2$$

For(int k=0; k<n; k++)

 $k=0 \Rightarrow 1$ $k<n \Rightarrow 1$ $k++ \Rightarrow 1$

$$n \times n = n^2$$

$$n \times n \times (n+1) = n^3 + n^2$$

$$n \times n \times n = n^3$$

System.out.println("Hello, world!") $\Rightarrow 1$

$$n^3 = n^3$$

+

$$\text{Total Freq} = 3n^3 + 4n^2 + 4n + 2$$

$$\text{Complexity} = O(n^3)$$

Question-4

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In this situation, I think the fastest way is Binary Search. Because we have infinite toys. This means the company needs to find the middle of the floors (50. floor). Then check if the toy is broken or not.

1- If the toy is broken this means that breaking point is between 1.-49. floors. So now the company should check middle of the first 50 floors (25.) Then again check if the toy is broken or not. If the broken the company should check the first part of the first 25 floors. And again and again. Until find the correct floor.

2- If the toy is Not broken in the first throw this means that breaking point is between 51-100. floors. So now the company should repeat same operations in case 1.

Complexity of the binary search is $O(\log N)$, we have 100 floors so our worst case is $\log(100)$, $\cong 7$ step. And our best case is 1. (in the first try) For this problem in the worst case, we will find the correct floor on the 7th try.